Computer Science Assignment
Topic: Fundamentals of Computer Science

Section 1: Operating Systems

1. Functions of an Operating System:
An operating system (OS) is a system software that manages hardware and software resources. Its prim
- Process Management
- Memory Management
- File System Management
- Device Management
- Security and Access Control

2. Difference between Process and Thread:
- A process is an independent execution unit that has its own address space, while a thread is a lightweig
- Threads share the same memory space, whereas processes do not.
- Processes require more time for creation and context switching compared to threads.

3. Scheduling Algorithms:
- First-Come, First-Served (FCFS)
- Shortest Job Next (SJN)
- Round Robin (RR)
- Priority Scheduling
- Multi-Level Queue Scheduling

4. Virtual Memory:
Virtual memory is a memory management technique that provides the illusion of a larger RAM. It enables
- Execution of large programs
- Process isolation
- Efficient memory allocation

5. Deadlock Prevention Techniques:
- Avoid circular wait
- Hold and wait prevention
- No preemption policy
- Mutual exclusion handling

Section 2: Computer Networks

1. OSI Model:
The OSI (Open Systems Interconnection) model consists of seven layers:
- Physical
- Data Link

- Network
- Transport
- Session
- Presentation
- Application

2. TCP vs. UDP:
   - TCP (Transmission Control Protocol) is connection-oriented, reliable, and ensures data integrity.
   - UDP (User Datagram Protocol) is connectionless, faster, and suitable for real-time applications.

3. Subnetting:
   Subnetting divides an IP network into smaller subnetworks, improving efficiency and security.

4. Network Topologies:
   - Bus
   - Star
   - Ring
   - Mesh
   - Hybrid

5. DHCP and DNS:
   - DHCP (Dynamic Host Configuration Protocol) assigns IP addresses dynamically.
   - DNS (Domain Name System) resolves domain names to IP addresses.

Section 3: Database Management Systems

1. ACID Properties:
   - Atomicity: Ensures complete transactions
   - Consistency: Maintains database integrity
   - Isolation: Prevents interference between transactions
   - Durability: Ensures data persistence

2. SQL vs. NoSQL:
   - SQL databases are structured and support ACID transactions.
   - NoSQL databases are flexible and handle large-scale data efficiently.

3. Normalization:
   A process to eliminate data redundancy and improve efficiency, categorized into:
   - 1NF (First Normal Form)
   - 2NF (Second Normal Form)
   - 3NF (Third Normal Form)
   - BCNF (Boyce-Codd Normal Form)

4. Indexing:
   Indexing speeds up data retrieval by creating a reference structure.

5. Transactions in DBMS:
   Transactions ensure data integrity through commit and rollback mechanisms.

Section 4: System Design

1. Scalability in System Design:
   Scalability refers to a system's ability to handle increased load efficiently.

2. Monolithic vs. Microservices Architecture:
   - Monolithic: Single-code base, tightly coupled components.
   - Microservices: Independent, modular services communicating via APIs.

3. Caching:
   Caching stores frequently accessed data to reduce latency and improve performance.

4. Load Balancing:
   Distributes network traffic across multiple servers to ensure stability and efficiency.

5. Database Sharding:
   Splits large databases into smaller partitions to improve performance and scalability.

Instructions:
- Answer all questions in detail.
- Provide diagrams where necessary.
- Submit the assignment as a PDF document.

Deadline: [Specify Date]